# Dear LLM, Can You Explain my MongoDB Database?

Applied Data Science - Project Proposal

# RESEARCH QUESTIONS & OBJECTIVE

**RQ1** How accurately can an LLM identify collections, relationships and modeling patterns from the JSON schema/data collection?

**RQ2** Are explanations grounded in concrete evidence and consistent with expert-labeled ground truth?



Starting from the JSON schema of a MongoDB database, design an LLM-based pipeline to automatically reconstruct:

- collections, fields and relationships
- modeling patterns: embedding, referencing, arrays, subset, bucket
- a reasoned, verifiable explanation of every choice

# IMPACT AND PRACTICAL VALUE

### Automatic documentation

Existing databases described in natural language, with relationships and design choices made explicit.

### Understand someone else's database

Quickly see how an existing database works and why it was built that way.

### Support for migrations and reviews

A checkable basis for refactoring, migrations and critical review of the data model.

### A benchmark for LLMs

Example databases labeled by experts, to measure how often an LLM gets patterns right.

# BASIC FLOW

LLM-based analysis

01

## Input

JSON schema, real documents

02

## Pattern recognition

Structure matched against operational definitions of each pattern

03

## Evidence & reasoning

Fields and documents supporting each choice, plus alternatives

04

## Output

Collections, relationships and modeling patterns, each with an evaluation

**EXAMPLE · orders collection**

```
{ _id: 1042,
  customerId: ObjectId("…"),
  items: [ { sku: "A12", qty: 2 },
           { sku: "B07", qty: 1 } ] }
```

**LLM OUTPUT**

`items[ ]` → **Embedding**

Confirmed

`customerId` → **Referencing**

Plausible

**Every attribution ends with an evaluation:**

| | |
|---|---|
| Confirmed | direct evidence in the data |
| Plausible | consistent, but not proven |
| Undeterminable | not enough information |

# LIGHT MENTORING

- Support during the laboratories of the course
- Off-line support via mail



Silvia Chiusano
silvia.chiusano@polito.it



Fiori Alessandro
alessandro.fiori@polito.it